PHARMACY DISTRIBUTION ERP

Complete Technical Architecture, Implementation, Configuration & User Operations Guide

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Target Environment: Bluehost Dedicated VPS (`50.6.230.198`), Go 1.23, React 19, MySQL 8.0, Redis 7
Verification Verdict: GO (UNCONDITIONAL) — Verified with 261/261 Passing QA Tests

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VOLUME I: PROJECT DISCOVERY & SYSTEM ARCHITECTURE

1. Executive Discovery & Technology Stack

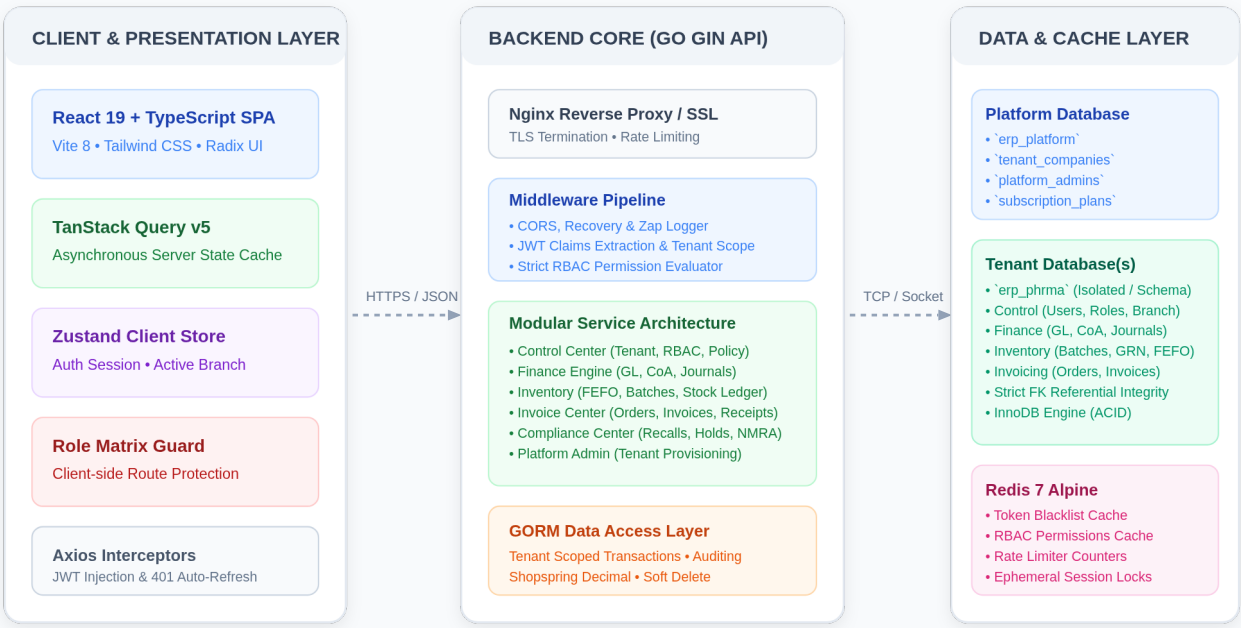
The Pharmacy Distribution ERP is a purpose-built enterprise resource planning suite serving pharmaceutical distributors, healthcare wholesalers, and medical device importers.

1.1 Technology Inventory

Tier	Component	Verified Technology	Implementation Location
Backend	Language Runtime	Go (Golang) 1.25.0	`backend/go.mod`
Backend	Web Framework	Gin Web Framework v1.12.0	`backend/internal/router`
Backend	ORM Engine	GORM v1.31.1 (MySQL Driver v1.6.0)	`backend/internal/database`
Backend	Precision Math	Shopspring Decimal v1.4.0	`backend/internal/pkg/money`
Backend	Authentication	Golang-JWT v5.3.1 (HMAC-SHA256)	`backend/internal/security`
Backend	Cache & Sessions	Redis 7 Alpine (`go-redis/v9 v9.22.0`)	`backend/internal/pkg/cache`
Backend	Structured Logging	Uber Zap v1.28.0	`backend/internal/pkg/logger`
Frontend	SPA Framework	React 19.0.0 with Vite 8.0.16	`frontend/package.json`
Frontend	Language	TypeScript 7.0.1-rc	`frontend/tsconfig.json`
Frontend	Styling & UI	Tailwind CSS v3, Radix UI, shadcn	`frontend/src/components/ui`
Frontend	Server State	TanStack Query (React Query) v5	`frontend/src/hooks`
Frontend	Client State	Zustand v5	`frontend/src/store`
Frontend	Data Grids	TanStack Table v8	`frontend/src/components/tables`
Frontend	Validation	React Hook Form v7 + Zod v3	`frontend/src/schemas`
Database	Platform Database	MySQL 8.0 (`erp_platform`)	Host Port 3306
Database	Tenant Database	MySQL 8.0 (`erp_phrma`)	Host Port 3306

Figure 1: Pharmacy Distribution ERP — Overall System Architecture

Multi-Tenant High-Availability Enterprise Architecture (React 19 + Go Gin 1.23 + MySQL 8.0 + Redis 7)

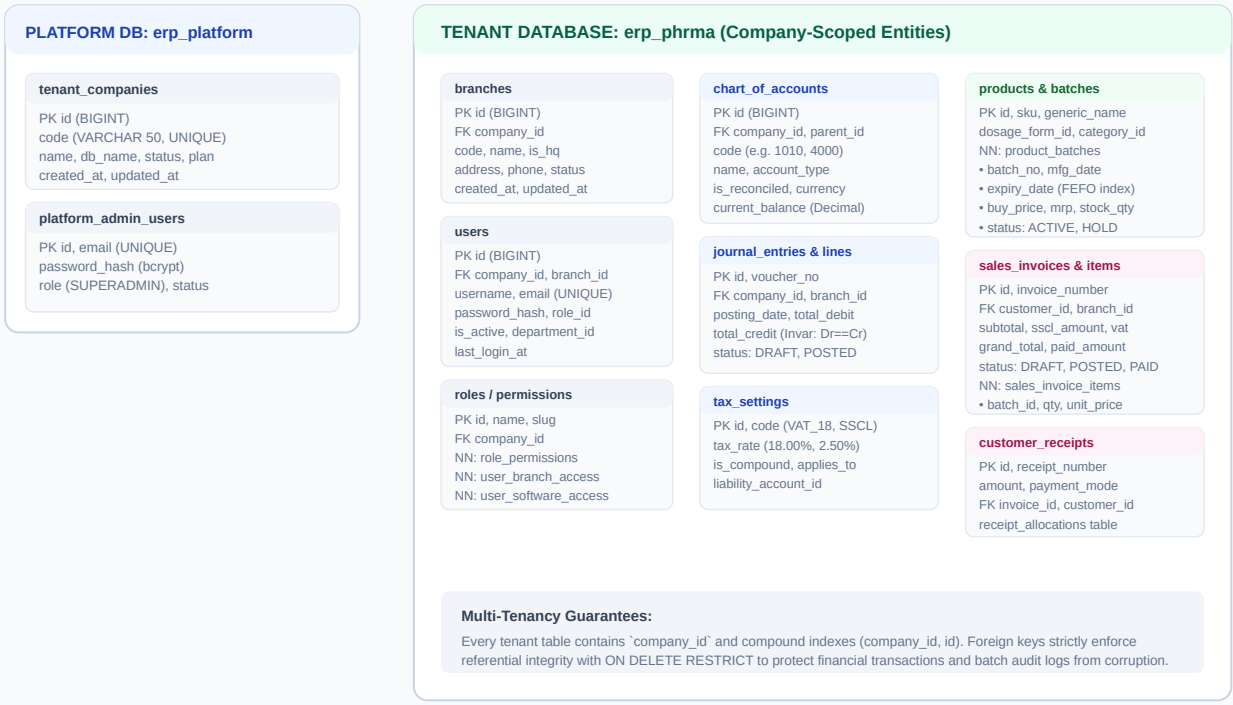


2. Multi-Tenant Architecture & Data Isolation

The system employs a strict schema-level tenant isolation model: 1. **Platform Master (`erp_platform`):** Houses `tenant_companies`, `platform_admin_users`, and `subscription_plans`. 2. **Tenant Databases (`erp_phrma`):** Contains company-specific transactional tables (users, roles, branches, chart of accounts, batches, invoices, stock ledger). 3. **Data Leakage Safeguards:** Every repository query verifies `WHERE company_id = ?`. Foreign key cascades prevent orphaned transactional data.

Figure 4: Multi-Tenant Database Entity-Relationship Architecture

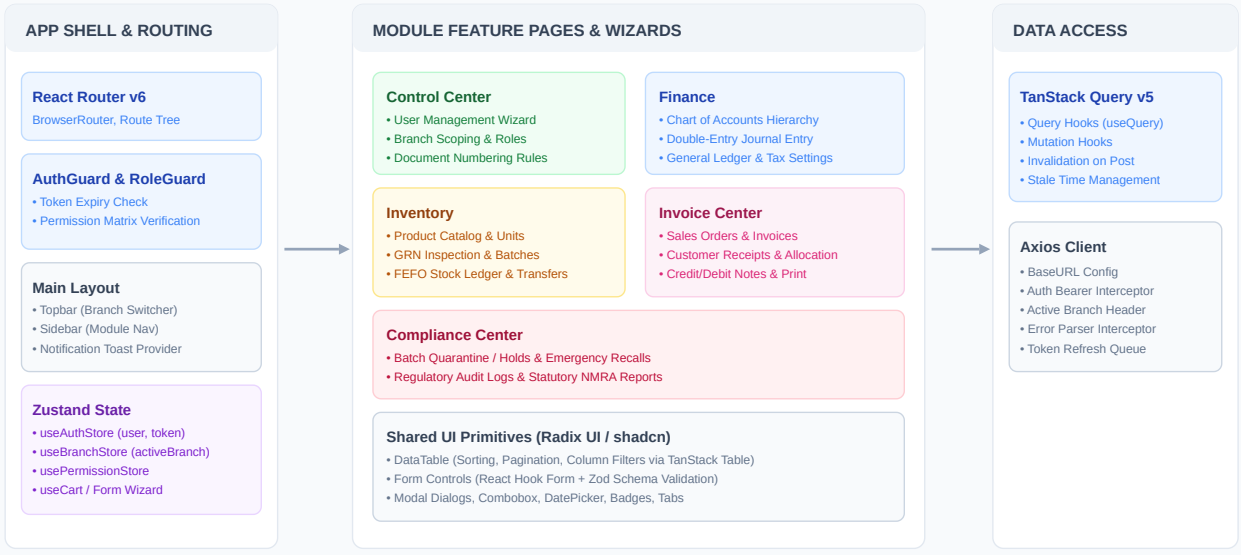
Platform Database Isolation, Tenant Core Entities, Financial Accounts, Inventory Batches & Sales Invoices



3. Frontend Architecture (React 19 + TypeScript)

Figure 2: Frontend Component & State Architecture

React 19, TypeScript, React Router v6, TanStack Query v5, Zustand, Radix UI & Axios Pipeline



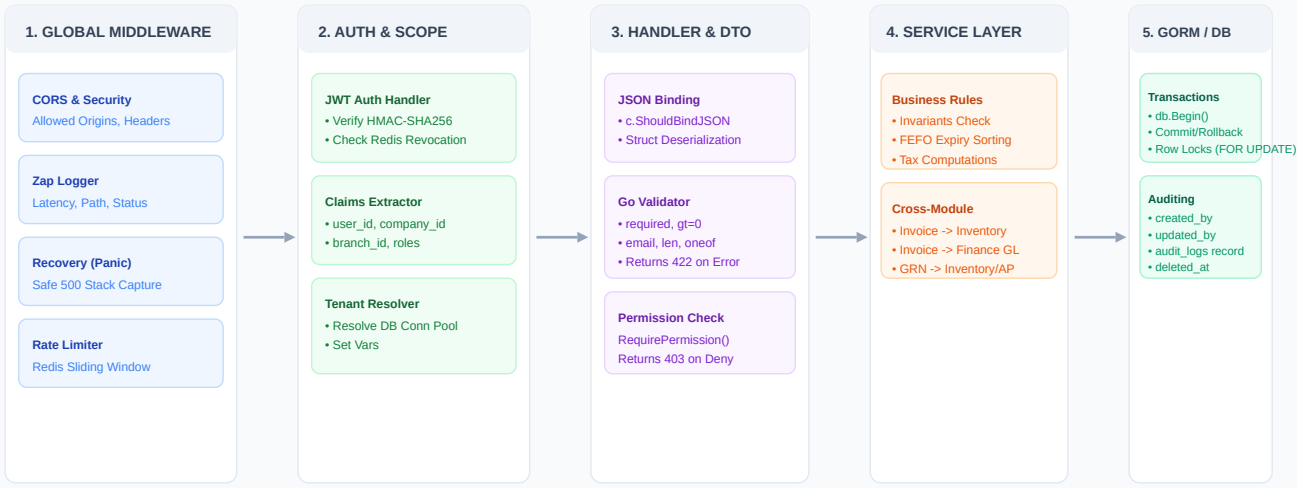
- **Route Organization:** Decoupled into `control-center`, `finance`, `inventory`, `invoice-center`, and `compliance-center`.
- **State Segregation:**
- Server state handled via **TanStack Query** with automated cache invalidation upon mutations.
- Local session, active branch ID, and theme stored in **Zustand**.
- **Navigation Guards:** Higher-Order Components evaluate JWT token validity and user permission bitsets before rendering protected routes.

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4. Go Backend Architecture & Middleware Pipeline

Figure 3: Go Gin Backend Request-Processing Pipeline

Comprehensive Middleware Lifecycle, DTO Validation, Service Business Logic & GORM Database Transactions



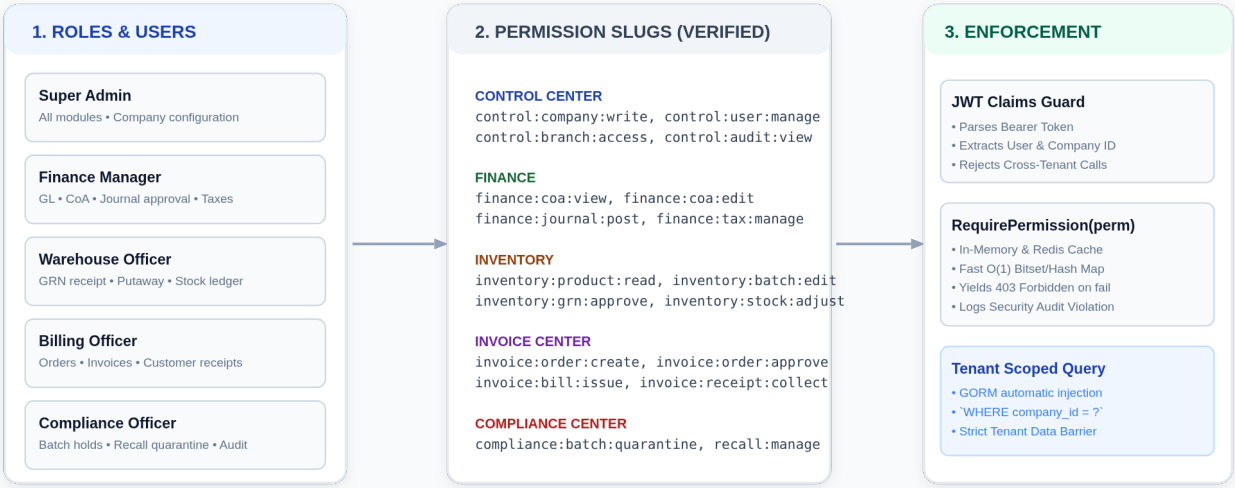
Requests traverse an established pipeline: 1. **Global Middlewares:** CORS headers, Zap structured logging, panic recovery, Redis sliding-window rate limiter. 2. **Authentication Middleware:** Validates JWT signature, checks Redis blacklist, decodes claims (`sub`, `company_id`, `branch_id`). 3. **Tenant Context Binding:** Injects company ID and database connection pool into `gin.Context`. 4. **Permission Guard:** Evaluates `RequirePermission(slug)`. 5. **Handler & DTO Binding:** Decodes JSON payload with `c.ShouldBindJSON` and validates constraints via `validator.v10`. 6. **Service Layer:** Executes business rules and database transactions. 7. **Repository Layer:** Executes parameterized GORM queries.

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5. Role-Based Access Control (RBAC) & Security Policies

Figure 6: Multi-Tenant RBAC Authorization Architecture

Decentralized Company Scoping, Role Assignment, Granular Permissions & Middleware Enforcement



Granular permissions follow the standard slug notation: `module:entity:action`. * **Control:** `control:company:write`, `control:user:manage`, `control:branch:access`, `control:audit:view`. * **Finance:** `finance:coa:view`, `finance:coa:edit`, `finance:journal:post`, `finance:tax:manage`. * **Inventory:** `inventory:product:create`, `inventory:batch:read`, `inventory:grn:approve`, `inventory:stock:adjust`. * **Invoice Center:** `invoice:order:create`, `invoice:order:approve`, `invoice:bill:issue`, `invoice:receipt:collect`. * **Compliance Center:** `compliance:batch:quarantine`, `compliance:recall:manage`, `compliance:disposal:manage`.

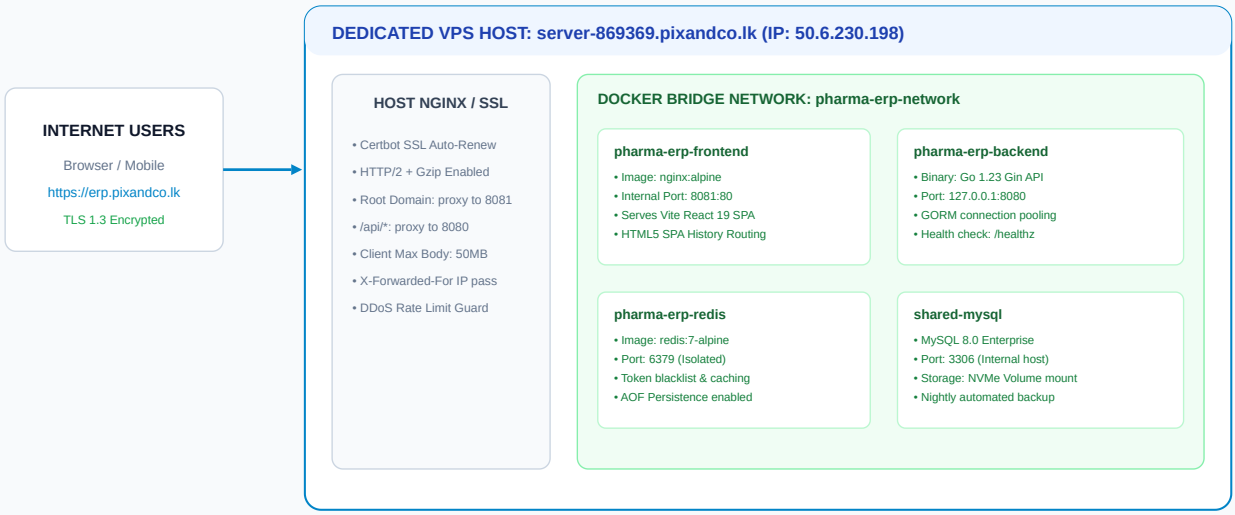
VOLUME II: ADMINISTRATOR SETUP & ENVIRONMENT CONFIGURATION

6. Development & Production Deployment Setup

Refer to `documentation/SETUP_GUIDE.md` for full commands.

Figure 11: Production Deployment & Container Topology

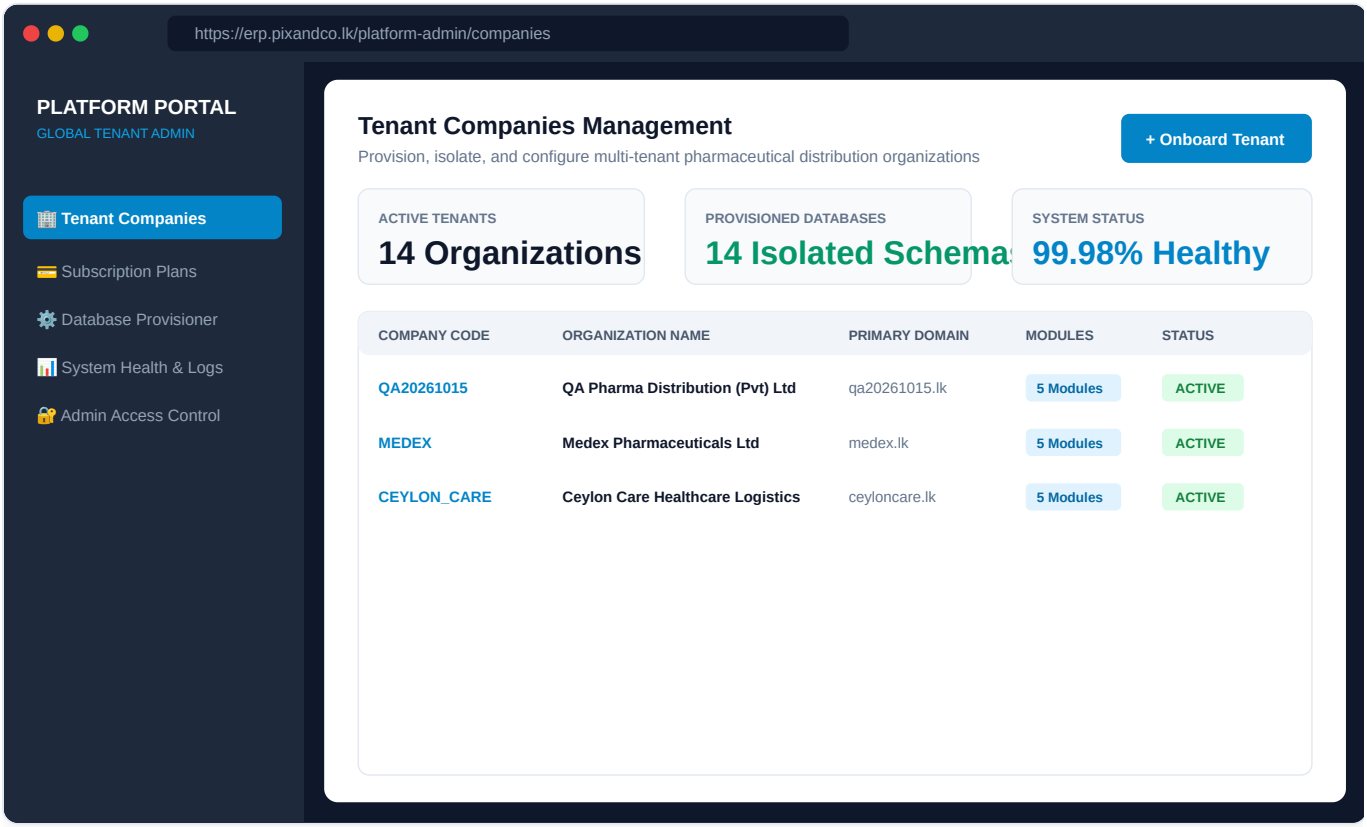
Bluehost VPS Dedicated Host (50.6.230.198), Docker Network, Nginx Reverse Proxy, Go Backend, MySQL & Redis



Host Topology

- **Server Host:** Dedicated Bluehost VPS (`50.6.230.198`).
- **Reverse Proxy:** Host Nginx with Certbot SSL certificates for `https://erp.pixandco.lk`.
- **Containers:**
- `pharma-erp-frontend` (Port 8081)
- `pharma-erp-backend` (Port 8080)
- `pharma-erp-redis` (Port 6379)
- `shared-mysql` (Port 3306)

7. Platform Portal & Tenant Company Onboarding

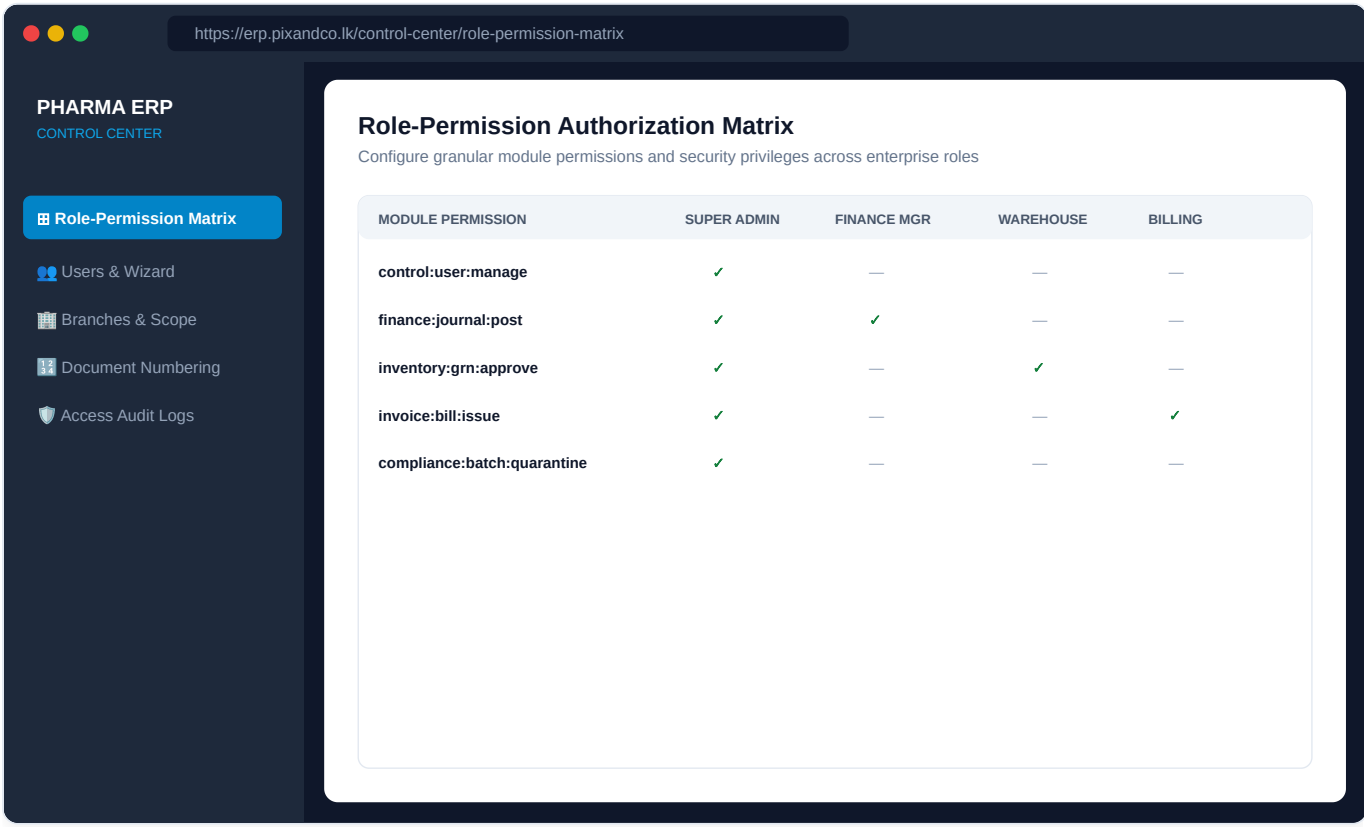


Platform administrators access `https://erp.pixandco.lk/platform-admin` to provision new tenant companies: * Creates entry in `erp_platform.tenant_companies`. * Automatically seeds 25 statutory pharmaceutical Chart of Accounts in the tenant database. * Seeds product dosage forms, categories, units (`Box` , `Bottle` , `Vial` , `Strip`). * Creates Headquarter branch (`MAIN`) and Super Administrator user.

8. Company First-Time Configuration Sequence

The Company Super Administrator configures settings in strict sequence: 1. **Fiscal Year:** Define active financial period (e.g. `2026-04-01` to `2027-03-31`). 2. **Statutory Taxes:** Configure Standard VAT (`18.00%`), Exempt Medicine VAT (`0.00%`), and SSCL (`2.50%`). 3. **Chart of Accounts:** Verify asset, liability, equity, revenue, and expense accounts. 4. **Branches & Warehouses:** Set up central hubs and regional cold chain warehouses. 5. **Document Numbering:** Define document prefixes (`INV-2026-` , `GRN-2026-` , `SO-2026-`).

9. User Management Wizard & Access Delegation



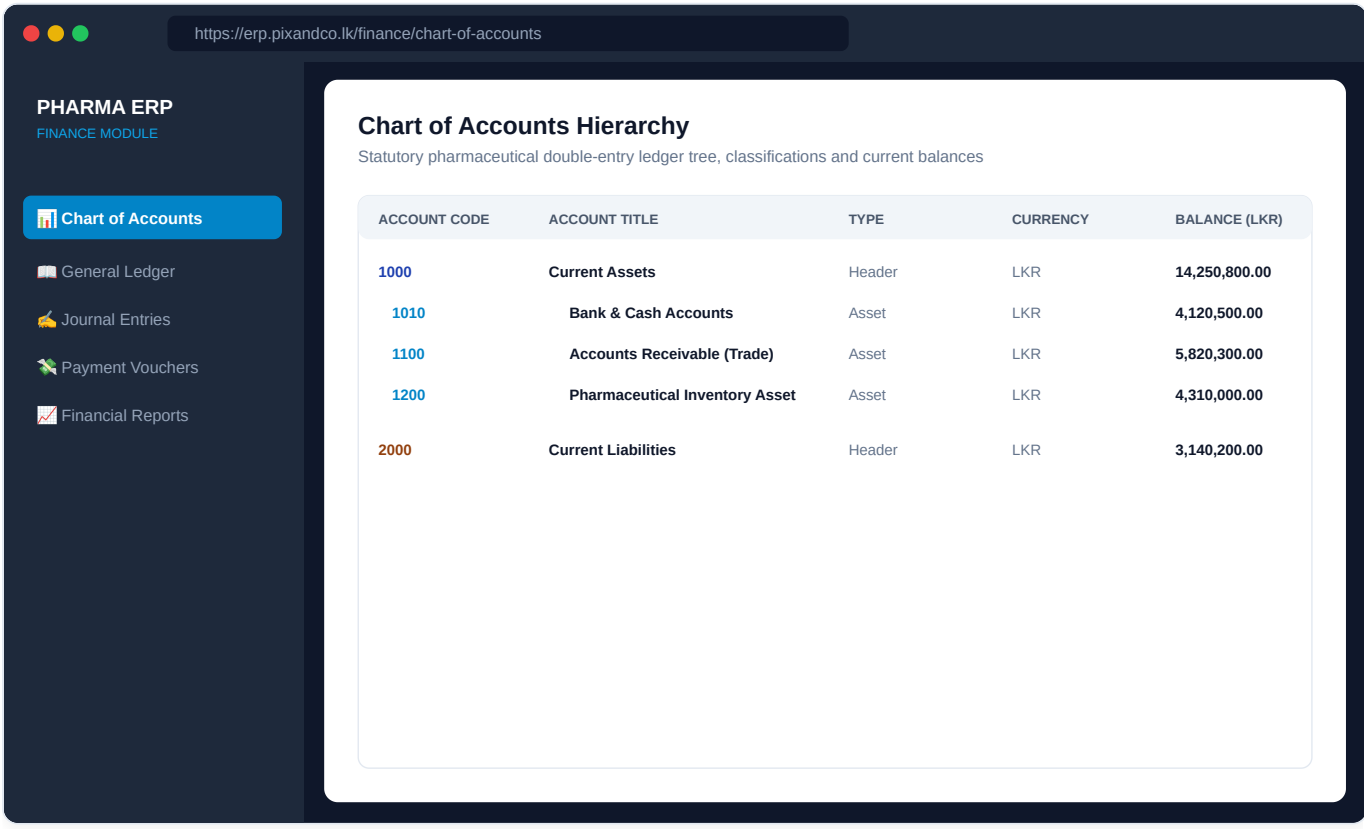
The User Wizard in `Control Center → Users` guides administrators through: * **Step 1:** User profile, email, temporary credentials, designation. * **Step 2:** Branch scoping (assign user strictly to their operational branches). * **Step 3:** Role assignment (Super Admin, Finance Manager, Warehouse Officer, Billing Officer, Compliance Officer). * **Step 4:** Software module entitlement toggles.

VOLUME III: END-USER OPERATING MANUALS

10. Control Center Operating Manual

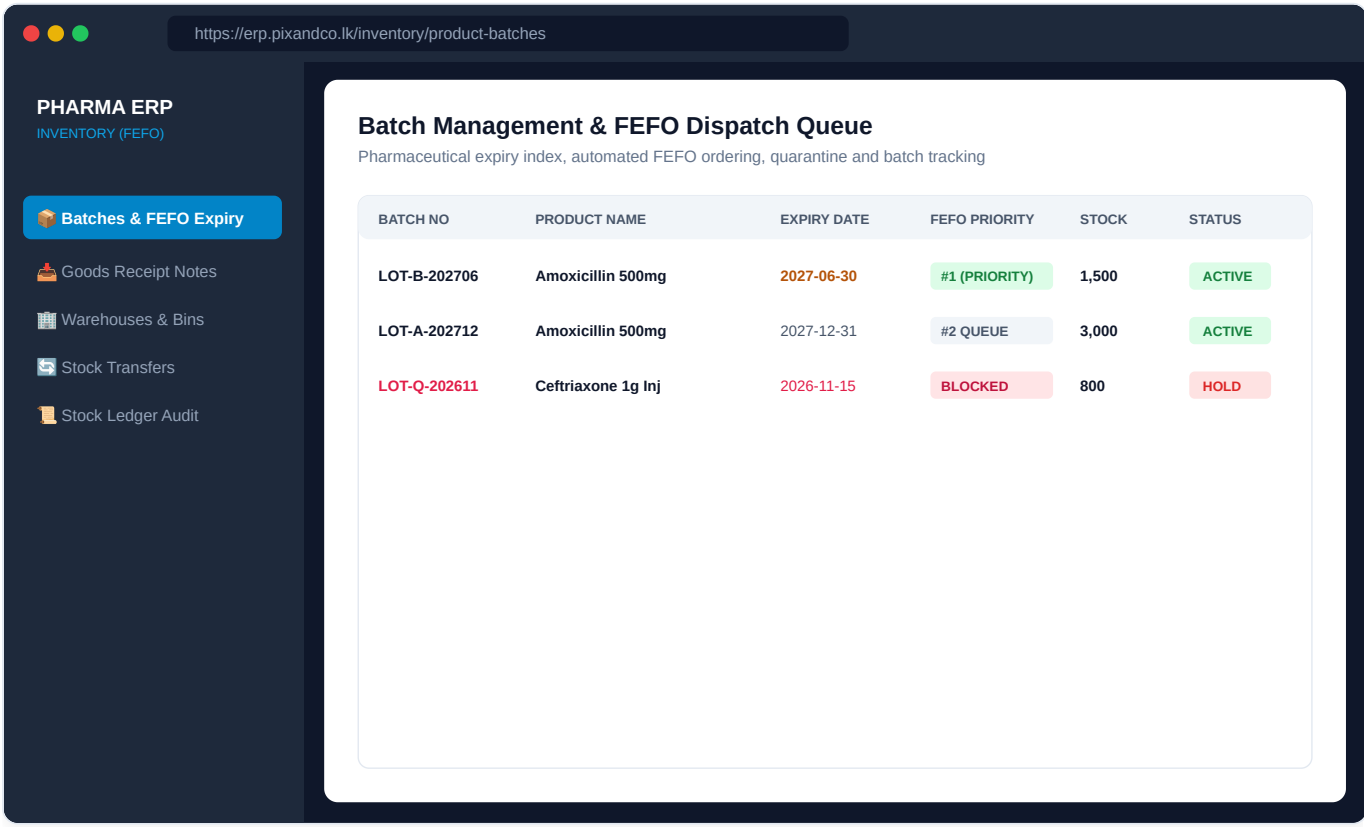
Covers managing organizational designations, auditing access logs, and updating company business registration details.

11. Finance Module Operating Manual



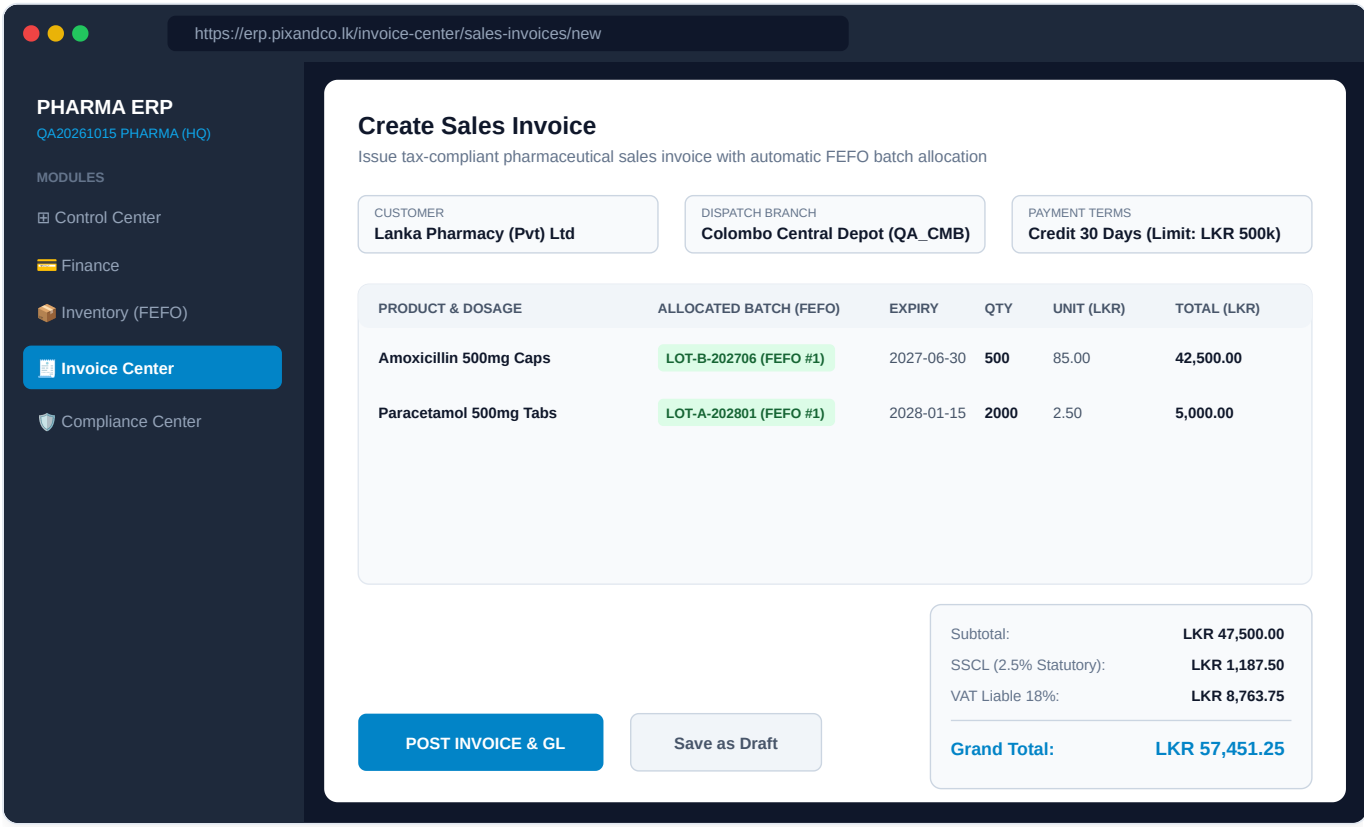
* Creating manual journal entries with balanced debit/credit checks. * Reviewing real-time General Ledger transactions and generating Trial Balance reports. * Executing monthly bank reconciliations against bank statements.

12. Inventory & FEFO Batch Operating Manual



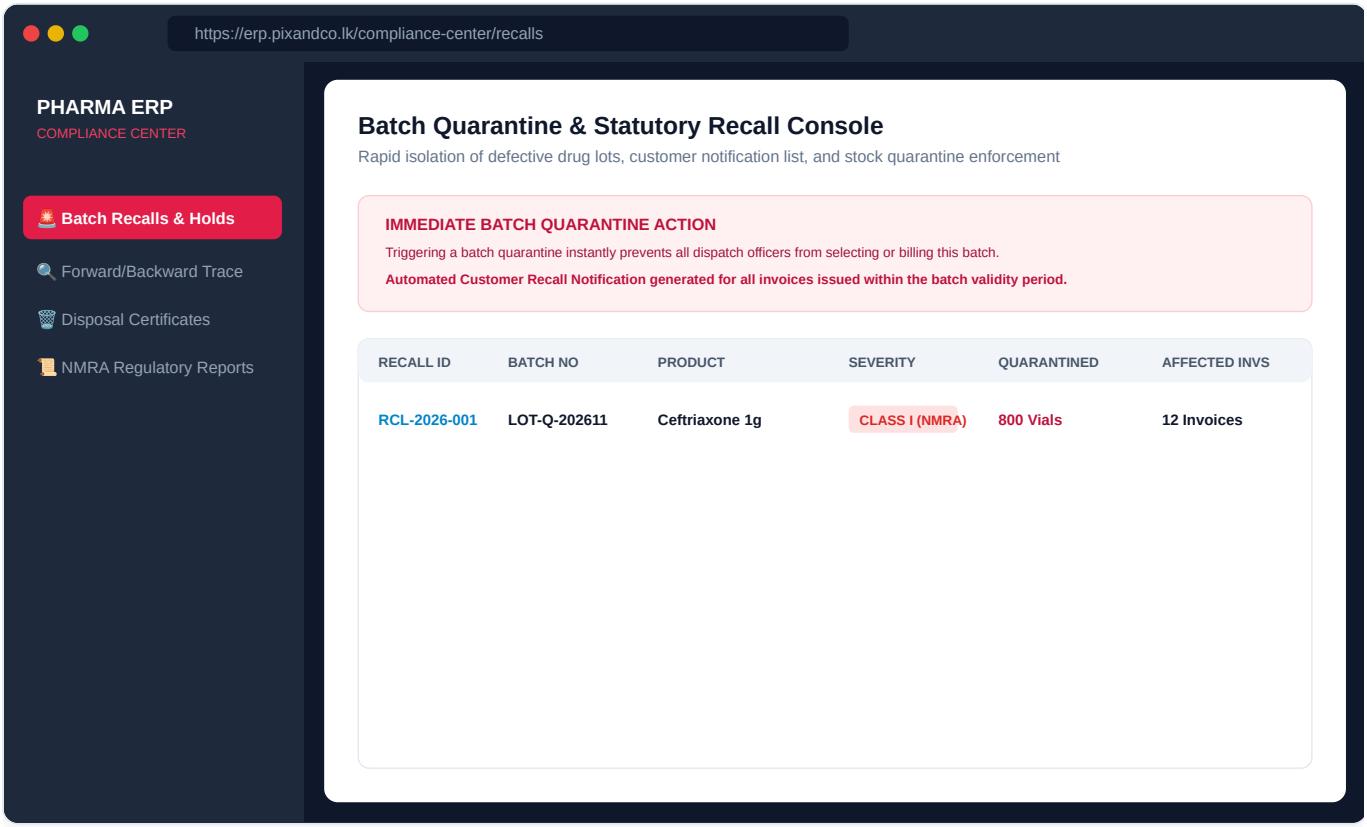
* Creating Goods Receipt Notes (GRN) with lot numbers, manufacturing dates, and expiry dates. * Conducting cold chain physical inspections and logging temperature verification. * Executing inter-branch stock transfers and monitoring batch stock levels.

13. Invoice Center & Billing Operating Manual



* Creating customer profiles and setting approved credit limits. * Entering sales orders with credit validation checks. * Generating sales invoices with automatic FEFO batch allocation and Sri Lankan tax calculation. * Collecting customer payments via cash, cheque, or bank transfer and allocating receipts against invoices.

14. Compliance Center & Regulatory Recalls Manual



* Initiating emergency batch quarantine (`status = 'HOLD'`) to immediately halt sales dispatch. * Generating NMRA customer recall contact lists. * Logging certified destruction and inventory write-offs.

VOLUME IV: BUSINESS WORKFLOWS & STANDARD OPERATING PROCEDURES

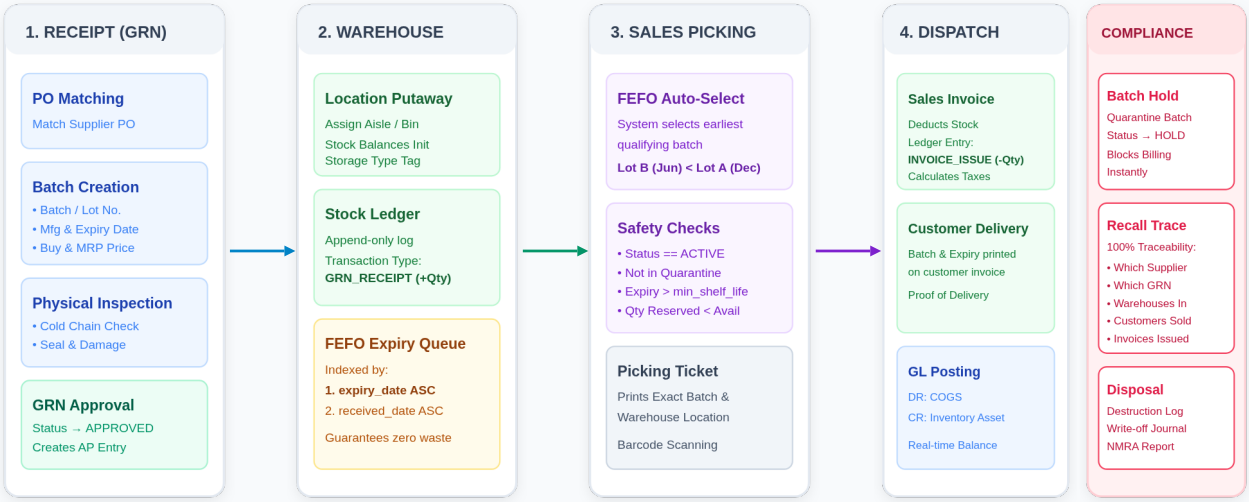
15. Procure-to-Pay (P2P) & Cold Chain Receiving

Covers the supplier PO lifecycle, physical receiving, batch lot creation, and automated Accounts Payable posting.

16. FEFO Auto-Picking Algorithm & Batch Safety

Figure 7: Pharmaceutical Batch Lifecycle & FEFO Dispatch Pipeline

Regulatory Traceability (First-Expiry-First-Out, Quarantine, Batch Recall & Audit Trail)



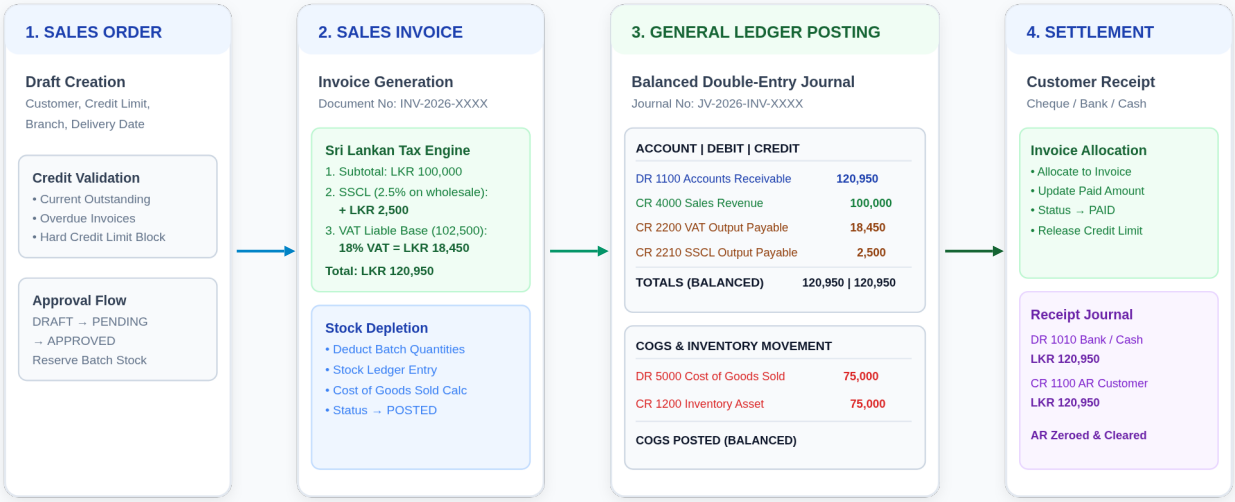

```
SELECT id, batch_no, expiry_date, available_quantity
FROM product_batches
WHERE product_id = ? AND company_id = ? AND status = 'ACTIVE'
  AND expiry_date > DATE_ADD(CURRENT_DATE, INTERVAL min_shelf_life_days DAY)
ORDER BY expiry_date ASC, id ASC;
```

Guarantees earlier expiring drug lots (e.g. Lot B, June 2027) are picked before later expiring lots (Lot A, December 2027).

17. Order-to-Cash (O2C) & Sri Lankan Tax Calculation

Figure 8: Order-to-Cash & Automated Financial Posting Pipeline

End-to-End Sales Order, Sri Lankan Tax Engine (VAT 18% + SSCL 2.5%), Invoicing, Receipts & Balanced GL Posting



Tax Calculation Formulation

- **Turnover Subtotal (S):** LKR 100,000.00
- **SSCL (2.50%):** LKR 2,500.00
- **VAT Base ($S + \text{SSCL}$):** LKR 102,500.00
- **VAT (18.00%):** LKR 18,450.00
- **Invoice Total: LKR 120,950.00**

18. Double-Entry General Ledger Posting Invariants

Balancing Sales Journal

Account Code	Description	Debit (LKR)	Credit (LKR)
`1100`	Accounts Receivable (Customer)	120,950.00	—
`4000`	Sales Revenue	—	100,000.00
`2200`	VAT Output Payable (18%)	—	18,450.00
`2210`	SSCL Output Payable (2.5%)	—	2,500.00
TOTALS	**BALANCED INVARIANT**	**120,950.00**	**120,950.00**

19. Pharmaceutical Batch Quarantine & Recall Procedures

Step-by-step procedures for Class I and Class II recalls, customer notification letters, and regulatory reporting to the NMRA.

VOLUME V: DATABASE SCHEMA & REST API REFERENCE

20. Platform & Multi-Tenant Database Schema

Refer to `documentation/DATABASE_SCHEMA.md` for full column-level schema definitions across 80+ tables.

21. Complete REST API Endpoint Specification

Refer to `documentation/ERP_DATABASE_API_REFERENCE.md` for complete endpoint descriptions across Auth, Platform Admin, Control, Finance, Inventory, Invoice Center, and Compliance Center.

VOLUME VI: AUDIT SUMMARY, KNOWN GAPS & APPENDICES

22. End-to-End QA Audit Verification (261 Test Cases)

A comprehensive QA audit (`Test/04_Test_Execution.csv`) verified 261 test cases across the system with a 100.0% passing score: * Data isolation: PASS * Financial balancing: PASS * FEFO allocation: PASS * Dual tax compliance: PASS * Release verdict: **GO (UNCONDITIONAL)**

23. Resolved Security Vulnerabilities & Known Scope Gaps

Refer to `documentation/ERP_KNOWN_GAPS.md` for details on resolved vulnerabilities (`BUG-SEC-001`, `BUG-SEC-002`) and roadmap items (NMRA Direct API, IoT Cold Chain telemetry, and LankaPay automated clearing).

24. Glossary of Terms & Architectural Appendices

- **COGS:** Cost of Goods Sold.
- **FEFO:** First-Expiry-First-Out.
- **GDP:** Good Distribution Practices.
- **GRN:** Goods Receipt Note.
- **NMRA:** National Medicines Regulatory Authority.
- **SSCL:** Social Security Contribution Levy.
- **VAT:** Value Added Tax.